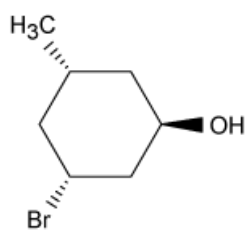


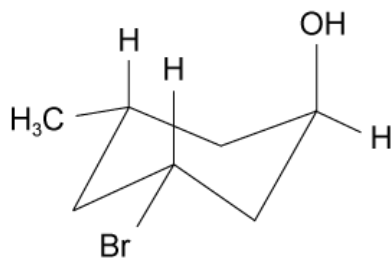
Compound 1 is shown below.



Compound 1

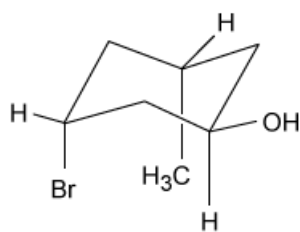
All of the following are equivalent to Compound 1 EXCEPT:

☐ A.



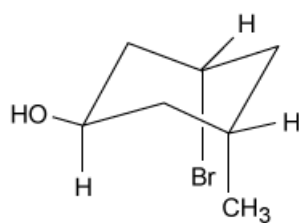
[12%]

☐ B.



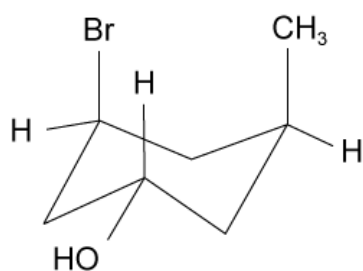
[8%]

☐ C.



[10%]

✓ ☒ D.

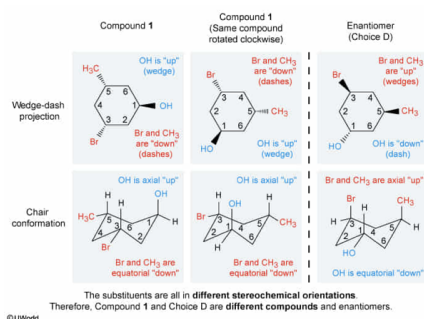


[68%]

Correct

68% answered correctly

Explanation:



Compounds can be illustrated in different ways and orientations. Molecules must have all substituents in the **same relative position** and **stereochemical orientation** to be considered the **same compound**. The cyclohexane ring, for example, can be depicted using the **wedge-dash projection** or a **chair conformation**. In the wedge-dash projection, the wedges indicate that the substituent is above or in front of the plane of the ring, and the dashes indicate that the substituent is below or behind the plane. In a chair conformation, the wedges and dashes can be shown as **axial** or **equatorial** positions.

Compound **1** is a six-membered ring shown in the wedge-dash projection with three substituents: -OH , -Br , and -CH_3 . The -OH group is bonded to carbon shown by a wedge, and -Br and -CH_3 are bonded to carbon atoms shown by dashed lines. The carbon atoms are numbered 1–6, beginning at the **alcoholic carbon** bonded to -OH and continuing in order to place -Br on carbon 3 and -CH_3 on carbon 5. For the orientations shown, this numbering is clockwise. The answer choices are chair conformations and must maintain the positions and stereochemical orientations of Compound **1** to be considered the same molecule. The **chair conformations** that depict -OH above the plane of the ring ("up") and -Br and -CH_3 below the plane ("down") and on carbons 3 and 5, respectively, are the same molecule as Compound **1** (**Choices A, B, and C**).

Choice D, however, positions the -OH substituent in an equatorial "down" position and the -Br and -CH_3 substituents in an axial "up" in this chair conformation, causing the stereochemical orientation in this structure to be **the complete opposite** (ie, mirror image) of Compound **1**. Rotation of the molecule to orient the -OH group "up" and the -Br and -CH_3 groups "down" would also switch the numbering of substituents to go from clockwise to counter-clockwise (ie, nonsuperimposable). Together, these features show that **Choice D** is a **nonsuperimposable mirror image** of Compound **1**, making them different compounds that are considered **enantiomers**.

Educational objective:

Compounds can be illustrated in different orientations and forms, including wedge-dash projections and chair conformations. For two depictions of a molecule to be considered the same compound, the atoms must have the same relative position and stereochemical orientation.

Time spent: 0 sec
Subject:
Organic Chemistry

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System:
Introduction to Organic Chemistry